Text-to-Image/Video Generation Models

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Text-to-image generation

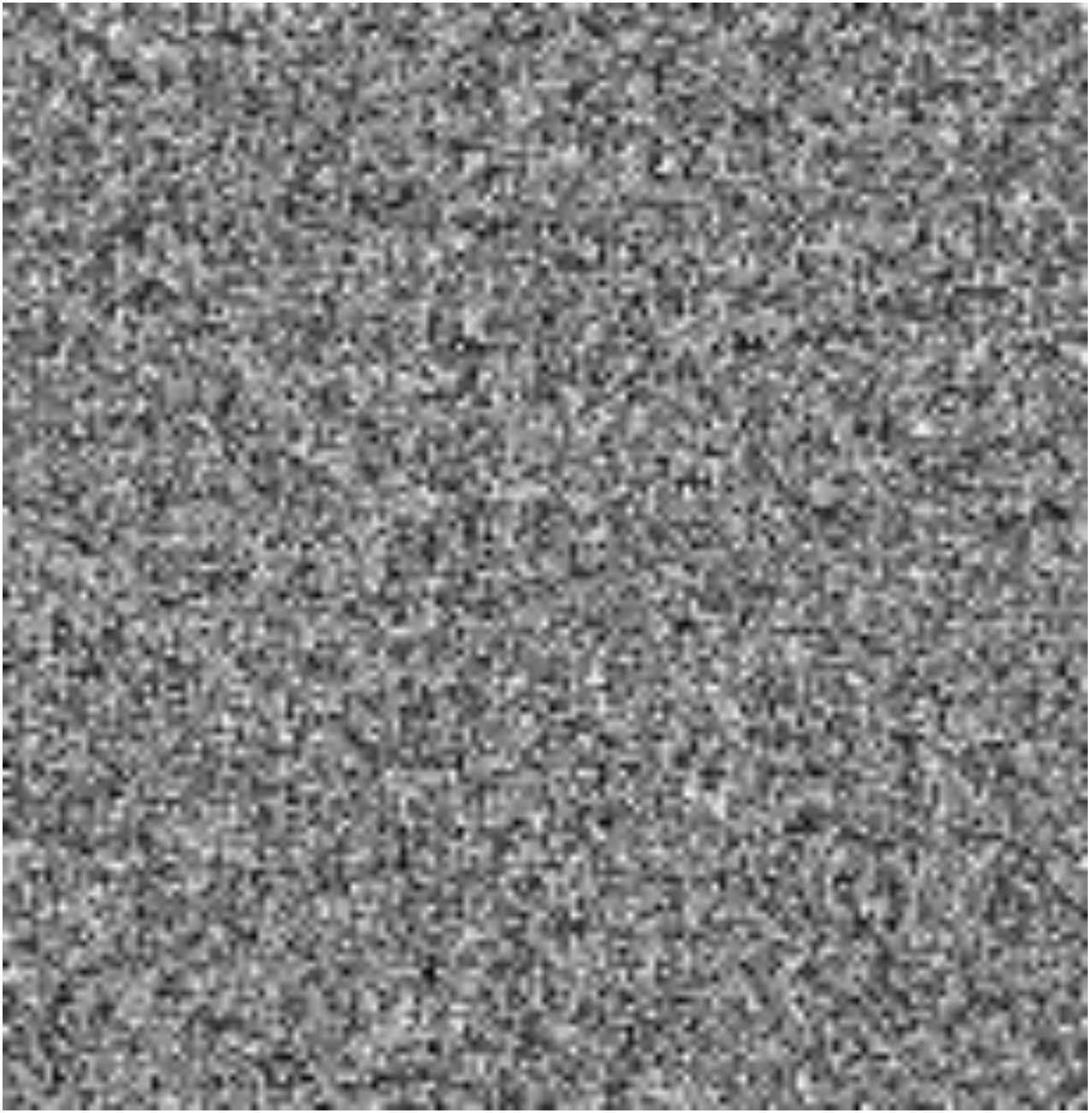
Text-to-video generation

***Text prompt**: A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick.*

Diffusion Models
(Stable Diffusion / Flux)

Video Diffusion Models (Sora2)

Implicitly learn text-image correspondence

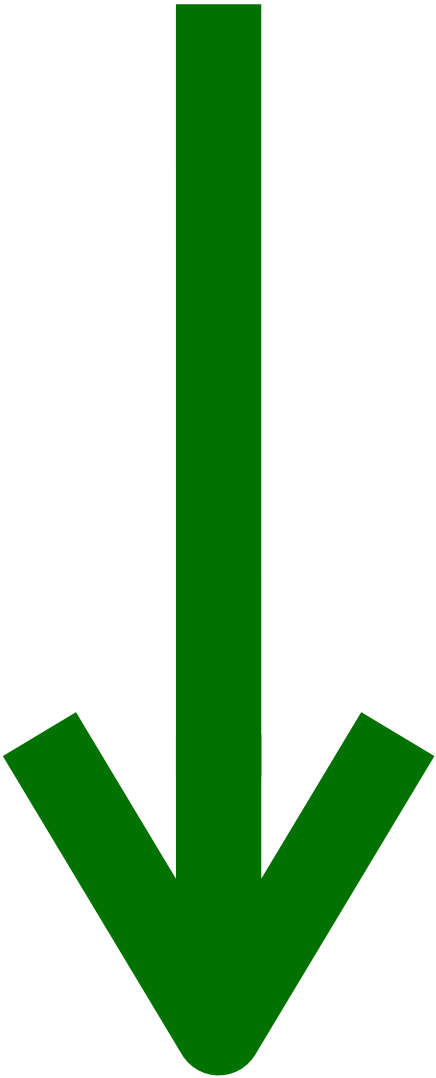


Noise

Output

Denoising process

Tiger



Text Guidance

city signage. She wears a black leather jacket, a

***Text prompt**: A stylish woman walks down a Tokyo*

purse. She wears sunglasses and red lipstick.

street filled with warm glowing neon and animated

long red dress, and black boots, and carries a black

(Stable Diffusion / Flux)

Diffusion Models

Video Diffusion Models

(Sora2)

Implicitly learn text-image correspondence

T2I Generation

(Noise → Image)

Cat

Training-free T2I Editing

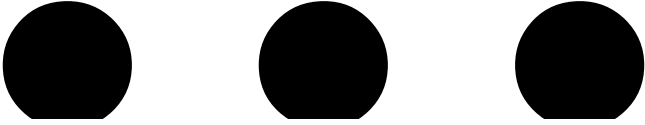
Diffusion Models (DMs) & Rectified Flow Models (RFs)

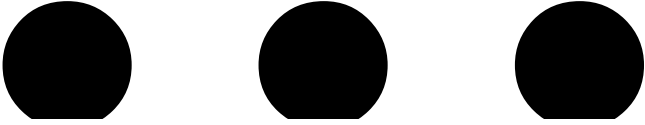
Training-Free Editing Based on Generation Models

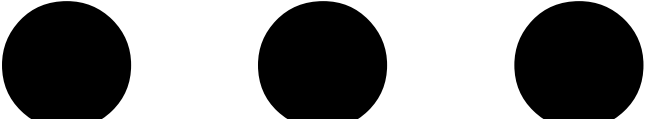
Noise → Image!

Modify generation trajectory:

Multiple time-step denoising process





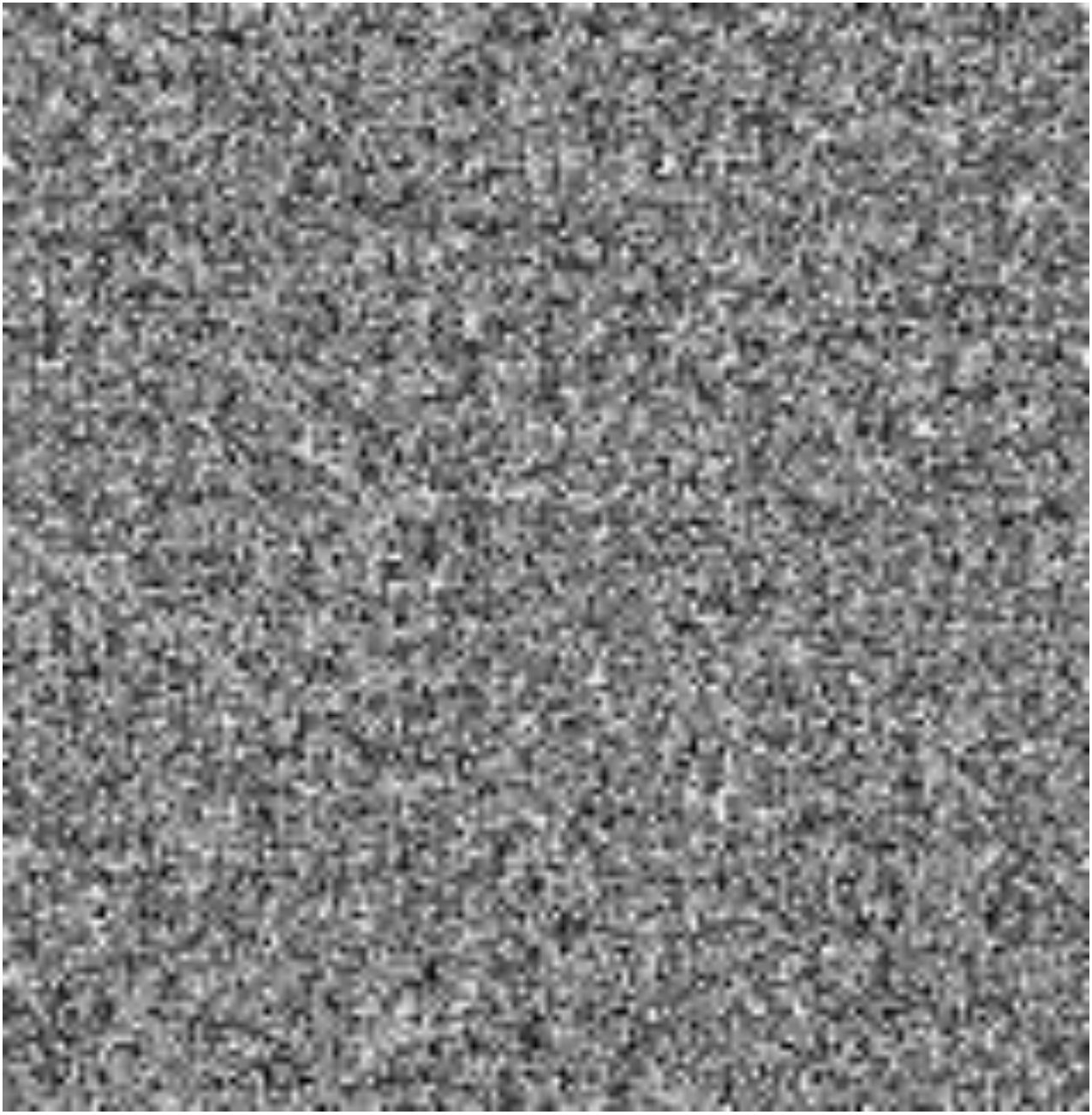


Noise

Output

Denoising process

Tiger



Noise

Output

Denoising process

Tiger

Text Guidance